

- Q.29 Differentiate between HV and LV transmission.
 Q.30 Explain importance of earthing in electrical systems.
 Q.31 Write applications of protective device like MCB and ELCB.
 Q.32 Define phase angle and its importance.
 Q.33 Explain working principle of an induction motor.
 Q.34 Compare AC and DC supply systems.
 Q.35 Define explosion proof motors and their uses.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain RMS, average peak and instantaneous values of AC with suitable equations and waveforms.
 Q.37 Describe three phase system in detail. Explain star and delta connections with advantages and applications.
 Q.38 Explain construction and working of PN junction diode. Discuss characteristics and applications of transistors.

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Roll No.

3rd Sem / Aircraft Maintenance Engg. Sub : Elements of Electrical And Electronic Engineering-I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the main purpose of measuring instruments in electrical systems?
 a) To increase circuit efficiency
 b) To indicate the magnitude of electrical quantities
 c) To reduce power loss
 d) To store electrical energy
- Q.2 Which instrument works on the principle of interaction between magnetic fields?
 a) Induction type b) Moving coil type
 c) Electrostatic type d) Thermocouple type
- Q.3 In a DC machine, what is the function of the commutator?
 a) To reduce voltage
 b) To convert AC to DC
 c) To increase current
 d) To store energy

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Q.4 In an alternator, which part produces rotating magnetic field?

- a) Stator b) Rotor
- c) Armature winding d) Slip rings

Q.5 What is the main function of a diode in a circuit?

- a) Amplification b) Rectification
- c) Oscillation d) Filtering

Q.6 What happens when a battery is excessively discharged?

- a) Voltage increase
- b) Battery life reduces
- c) Resistance becomes zero
- d) Temperature decreases

Q.7 Which filter allows only a specific range of frequencies to pass?

- a) Low pass filter b) High pass filter
- c) Band pass filter d) Notch filter

Q.8 If the inductance of a coil increases, what happens to reactance (for constant frequency)?

- a) Decreases b) Increases
- c) Remains same d) Becomes zero

Q.9 The energy stored in a capacitor is of _____ nature.

- a) Magnetic b) Electrostatic
- c) Mechanical d) Thermal

Q.10 At resonance in an RLC circuit impedance is:

- a) Maximum b) Minimum
- c) Infinite d) Zero

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define alternating current.

Q.12 What is electrical potential difference?

Q.13 What is the working principle of a transformer?

Q.14 Define time period of an AC waveform.

Q.15 What is peak value of AC?

Q.16 What is meant by step up transformer?

Q.17 Why is high voltage used in power transmission?

Q.18 What is the function of a fuse?

Q.19 What is isolation in electrical systems?

Q.20 Define battery efficiency.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain methods of improving power factor.

Q.22 List advantages of electrical energy.

Q.23 Define resistance, inductance and capacitance.

Q.24 Compare single phase and three phase systems.

Q.25 Explain form factor and its significance.

Q.26 Describe working of a single-phase transformer.

Q.27 Explain star and delta configurations with diagrams.

Q.28 Discuss types and applications of batteries.